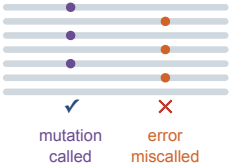
A Where the background comes from

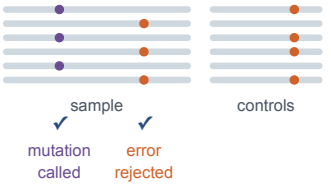
Read-level test (e.g. LoFreq, Geneious)



Each base is judged only by its own quality score. Sequencing errors that recur at one position (PCR, homopolymers, paralog reads) pass this test and are wrongly counted as SHM mutations, and reads from one animal are treated as independent replicates.

✓ correct decision ✕ wrong decision

Control-calibrated test (SHMtrace)



Error rates per substitution, strand and base quality are measured in matched AID-null libraries from the same run. A call needs support beyond this background (FDR ≤ 0.05), and groups are compared with one value per animal.

B Inputs and read preparation



Nanopore
full-length amplicons



Sanger
.ab1 traces + sample sheet



Illumina
paired reads, merged



UMI-tagged
one consensus per molecule

Read preparation

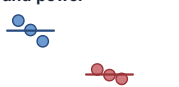
1. **Length and expected-error gates**
   over-length reads and reads averaging > 0.01 errors per base
2. **Competitive alignment**
   minimap2 against target and paralog references; chimeras out
3. **Paralog-allele filter**
   reads with ≥ 10 paralog-specific bases removed
4. **Matched depth**
   random draw of the same number of usable reads per library

C Calling and quality checks

1. **Control error model**
   per substitution, strand and base quality; a margin chosen by leave-one-control-out
2. **Test and FDR**
   support against expected errors; FDR ≤ 0.05 set from the controls; controls tested against each other
3. **Germline and window**
   sites at ≥ 50% removed; calls restricted to the covered window; indels optional
4. **Automatic checks**
   spectrum (C→T/G→A share), positional clustering, 5-mer targeting index, Sanger bookkeeping flags

D Outputs

Animal-level tests and power
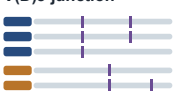


NB, Welch, permutation

Single-read estimate



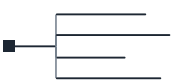
observed minus control background

Clones from the V(D)J junction



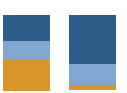
reads grouped by junction; each clone counted once

Mutation trees



full-length reads of one clone

Repair-route indices



G:C Ts / Tv, A:T, WA/TW, decomposition

Reports, app and command line



HTML report, tables, project files